



July 2015

Team: SC, RM, LC (75%), NW (20%)

Strategy

- [LC] Research and reporting on landscape of primary mortgage market
 - Added Mortgage Credit Availability Index
 - Market concentration report for Ginnie Mae
- [RM, LC] Productionized first version of proprietary Mortgage Credit Availability (MCA) measure by running the most recent originations through Accordion each month and tracking a default measure. Presented model to Investment Committee.
- [SC, LC] CRE Business Launch: Presentation framework for CRE Equity L/S strategy.
- [SC] Slides and write-up for Quarterly call

Modeling

- **Scale:** Reorganize and refactor existing CCMQuantlib fixed income analytics library (written by Bin Shao) as Scale. See the Scale presentation (distributed separately) for an overarching view of the goals of this library.
 - [JH, NW, RM] New Scale: Work on productionizing Quantlib code that replicates Bin's code. If successful, this approach would provide an excellent platform for our Scale project. Quantlib is:
 - Reasonably well-tested: it's been around since 2000
 - Well-maintained: There are at least a couple of releases every year
 - Rich feature set: Quantlib functionality includes a significant collection of utilities (Date libraries, optimization engines, splines etc.), yield curve builders, interest rate models etc.
 - Documented code
 - [NW] Building a validation check to check estimated LIBOR/Swap rates against inputs when QL fails to complete this check. LIBOR is complete.
- **Baton** [RM]: Agency prepayment model
 - Presentation of turnover sub-model along with model tracking results to investment committee
- **Accordion** [RM, LC]: Roll-rate model for GSE loan-level data



- First version of deal-level surveillance model for CRT deals with eMBS CRT data [LC]
- Automate running CRT deals through Accordion using eMBS CRT data on a monthly basis [RM]



August 2015 Pipeline

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Strategy

- [SC] Refinement of IOS Basis Model
- [SC, LC] CRE Strategy: Continue to flesh out Investment Process and Risk Management presentations & strategy for CRE L/S Equity and CMBS/CMBX
- [SC, LC, ES, JE] Revamp Investor Presentation for Fixed Income Master Fund
- [SC] Steve Gallant business plan: assess

Modeling

- **Scale** [NW] Write native code instead of relying on QuantLib itself for checking results produced by Yield Curve Builder. [RM] Add logging structure to Scale Prepayment Model. [NW, RM] Add Unit Test framework to Yield Curve Builder, Prepayment Model.
 - Present Scale progress report to Investment Committee [NW]
- **Accordion** [RM, LC]: Roll-rate model for GSE loan-level data
 - Add tracking data to surveillance report process [RM, LC]
 - Finish surveillance report with revisions and productionize on a monthly basis [LC]
 - Second iteration of Cello CRT primer [LC, RM]
 - Loss severity model [RM, LC]
 - Integrate Accordion forecasts with CRT Risk Sheet [RM, LC, JH]
- **Baton** [RM]: Agency prepayment-model
 - Refinance sub-model
 - Documentation of modeling assumptions and organization of work done so far.

Analytics

- **Cello Grid Computing Plan** [DH, JH, RM, SC] Formulate strategy